

Gio Jung

Machine Learning Engineer, Multimodal ML Systems, LLM Evaluation
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Experience

- **Research Assistant, YouDescribe Lab** Jun 2024 - Present
Summary: Accessibility research lab advancing video audio description (AD) for blind and low-vision users. Leading systematic model evaluation research to validate the performance of VLM as rater and describer.
 - Leading key frame selection research to improve VLM-based evaluation and description of video AD.
 - Validated 100+ VLMs as raters across image captioning and video AD by building end-to-end evaluation pipelines with Hugging Face, LangChain-based API integrations supporting 2 ongoing research submissions.
 - Uncovered model-specific strengths and failure patterns across 7 quality rating dimensions by independently applying Item Response Theory (IRT) frameworks to compare VLM and human rater performance across task domains.
 - **SpaceTimeGPT** 2025
Summary: NAIRR Pilot-funded project scaling a video captioning model (TimeSformer $\times$ GPT-2).
 - Prepared and transferred 60K+ VATEX video clips to Pittsburgh Supercomputing Center (PSC) HPC clusters, preprocessing frames for efficient distributed data loading at scale.
 - Co-trained a 400M+ parameter video captioning model across 16 NVIDIA H100 GPUs (2 nodes) using DeepSpeed Pipeline Parallelism, achieving CIDEr 63 on the VATEX benchmark.
- **Software Engineering Intern, Astro Information Security** Aug 2025 - Nov 2025
Summary: Information security startup. As an intern, I built internal email automation tooling.
 - Designed an email classification MCP pipeline to process 1K+ messages with automated rule-based filtering.

Projects

- **eBay 2025 University ML Competition** 2025
 - Ranked Top 4 by iteratively fine-tuning sequence labeling models for a German marketplace, reaching $F1 = 0.94$ through systematic preprocessing and model selection.
- **ChillMate** Fall 2024
 - Built an AI mental health chatbot to make emotional support more accessible, using React, MongoDB, and a RAG-powered conversation engine, while leading a team of 7 as Scrum Master through full project delivery.

Publications

- Juvenal Francisco Barajas*, **Gio Jung***, Manali Seth, Lana Do, Malleeswari Jagabattuni, Andrew Scott, Shasta Ihorn, Alexander M. Blum, Vassilis Athitsos, and Ilmi Yoon. *Beyond Lexical Overlap: Validating Semantic Similarity Metrics for LLM-Based Video Caption Augmentation*. In Press at ICIIT 26.
- Alexander M. Blum, James M. Mason, Robin C. Irey, Yukie Toyama, Yunting Liu, **Gio Jung**, Andrew Scott, JinHo Kim, David Pearson. *Disentangling local versus global processing dispositions in autistic cognition using item response theory models: comparing multimodal comics versus text-based narratives in a randomized study*. Journal of Discourse Processes.

Education

San Francisco State University

- M.S. in Computer Science (2025); B.A. in Mathematics, Minor in Computer Science (2023)
- Tutor, TASC; Grader/TA, Dept. of Mathematics

Awards & Grants

NAIRR Pilot | National Science Foundation (NSF) | Google CAHSI IRP Scholar | Jessie F. L. York Scholarship

Technical Skills

- **Programming & Data:** Python, SQL, Bash, pandas, NumPy, data curation, batch inference
- **ML & Multimodal Systems:** PyTorch, Hugging Face Transformers, multimodal pipelines, fine-tuning, RAG, APIs
- **Evaluation & MLOps:** IRT, Kendall's tau, BLEU, CIDEr, METEOR, CLIPScore, frame sampling, DeepSpeed, W&B